

Reviewing the *Clostridioides difficile* Mouse Model: Insights into Infection Mechanisms.

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Clostridioides difficile is an anaerobic, spore-forming bacterium associated with intestinal infection, manifesting a broad spectrum of gastrointestinal symptoms, ranging from mild diarrhea to severe colitis. A primary risk factor for the development of *C. difficile* infection (CDI) is antibiotic exposure. Elderly and immunocompromised individuals are particularly vulnerable to CDI. A pivotal aspect for comprehending the complexities of this infection relies on the utilization of experimental models that mimic human CDI transmission, pathogenesis, and progression. These models offer invaluable insights into host-pathogen interactions and disease dynamics, and serve as essential tools for testing potential therapeutic approaches. In this review, we examine the animal model for CDI and delineate the stages of infection, with a specific focus on mice. Our objective is to offer an updated description of experimental models employed in the study of CDI, emphasizing both their strengths and limitations.

Introduction to the Special Issue on *Clostridioides difficile*.

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The Gram-positive, anaerobic bacterium *Clostridioides difficile* (CD) represents the most common cause of nosocomial diarrhea worldwide and is responsible for increased morbidity and mortality, and prolonged hospital stays [...].

Establishment of a gnotobiotic pig model of *Clostridioides difficile* infection and disease.

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Clostridioides difficile (*C. difficile*) is a gram-positive, spore-forming, anaerobic bacterium known to be the most common cause of hospital-acquired and antibiotic-associated diarrhea. *C. difficile* infection rates are on the rise worldwide and treatment options are limited, indicating a clear need for novel therapeutics. Gnotobiotic piglets are an excellent model to reproduce the acute pseudomembranous colitis (PMC) caused by *C. difficile* due to their physiological similarities to humans and high susceptibility to infection. Here, we established a gnotobiotic pig model of *C. difficile* infection and disease using a hypervirulent strain. *C. difficile*-infected pigs displayed classic signs of *C. difficile* infection, including severe diarrhea and weight loss. Inoculated pigs had severe gross and microscopic intestinal lesions. *C. difficile* infection caused an increase in pro-inflammatory cytokines in samples of serum, large intestinal contents, and pleural effusion. *C. difficile* spores and toxins were detected in the feces of inoculated animals as tested by anaerobic culture and cytotoxicity assays. Successful establishment of this model is key for future work as therapeutics can be evaluated in an environment that accurately mimics what happens in humans. The model is especially suitable for evaluating potential prophylactics and therapeutics, including vaccines and passive immune strategies.

Suppressive effect of *Lactobacillus fermentum* Lim2 on *Clostridioides difficile* 027 toxin production.

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Clostridioides difficile is a spore-forming, Gram-positive, anaerobic pathogen that caused gastrointestinal illness. During dysbiosis, overgrowth of *C. difficile* resulting in higher levels of toxin production. Since *Lactobacillus* has been commonly used to alleviate gastrointestinal discomfort, this study aimed to investigate the effects of *Lactobacillus* isolated from kimchi on the quorum-sensing and virulence factors of *C. difficile* 027. Among the isolated *Lactobacillus* strains, the acid and bile tolerant *L. fermentum* Lim2 was only able to reduce *C. difficile* 027 growth by one log₁₀ CFU per ml. In keeping with this finding, *C. difficile* 027 growth was unaffected by either untreated or heat-inactivated cell extracts from *L. fermentum* Lim2. Both untreated and heat-inactivated cell extracts did, however, significantly reduce the autoinducer-2 (AI-2) activity of *C. difficile* 027, with the most prominent suppression effect (654-fold) being found from 100 mg ml⁻¹ of heat-inactivated cell extract. A gene expression analysis indicated that in the presence of 100 mg ml⁻¹ heat-inactivated cell extract, the quorum-sensing (*luxS*) and the virulence factors (*tcdA*, *tcdB* and *tcdE*) were significantly suppressed, whereas the negative regulator gene (*tcdC*) was significantly up-regulated. Taken together, the significant anti-pathogenic effect from *L. fermentum* Lim2 could potentially be used to treat *C. difficile*-infections. SIGNIFICANCE AND IMPACT OF THE STUDY: *Clostridioides difficile* is a Gram-positive pathogenic bacteria that caused gastrointestinal illness via toxic production. The emergence of highly virulence and foodborne *C. difficile* strains has further increased the incident and severity of *C. difficile*-infections (CDIs). Numerous studies have reported the immunomodulatory activity of *Lactobacillus*, a member of healthy gut microbiota, to maintain gastrointestinal health. Here, we successfully isolated *L. fermentum* Lim2 from kimchi, and identified a promising anti-pathogenic effect against *C. difficile* 027, from the heat-inactivated *L. fermentum* cell extract via suppression on the *C. difficile* 027 quorum-sensing system and toxin production, which could potentially be used to treat and prevent CDIs.

Diverse Sources and Latent Reservoirs of Community-Associated *Clostridioides difficile* Infection.

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Clostridioides difficile is a spore-forming, toxin-producing, anaerobic bacterium that infects the human gastrointestinal tract, causing diarrhea and life-threatening colitis. *Clostridioides difficile* epidemiology continues to evolve, and it is recognized as a major community-associated (CA) pathogen in addition to its established role in causing healthcare-associated (HA) infection. While current surveillance and prevention measures mainly focus on healthcare-associated *C. difficile* infections (HA-CDI), much less is known about the factors that drive CA-CDI. This review highlights the potential contribution of reservoirs, including asymptomatic carriers, to CA *C. difficile* transmission. The reservoirs discussed in this review provide potential avenues for research to better understand and reduce CA transmission of *C. difficile*.

Identification of linear epitopes on the flagellar proteins of *Clostridioides difficile*.

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Clostridioides difficile (*C. difficile*) is an opportunistic anaerobic bacterium that causes severe diseases of the digestive tract of humans and animals. One of the possible methods of preventing *C. difficile* infection is to develop a vaccine. The most promising candidates for vaccine antigens are the proteins involved in

the adhesion phenomena. Among them, the FliC and FliD are considered to be suitable candidates. In this paper, the FliC and FliD protein polypeptide epitopes were mapped in silico and by using PEPSCAN procedure. We identified four promising epitopes: 117QRMRTLS123, 205MSKAG209 of FliC and 226NKVAS230, 306TTKKPKD312 of FliD protein. We showed that 117QRMRTLS123 sequence is not only located in TLR5-binding and activating region, as previously shown, but forms an epitope recognized by *C. difficile*-infected patients' antibodies. 205MSKAG209 is a *C. difficile*-unique, immunogenic sequence that forms an exposed epitope on the polymerized flagella structure which makes it a suitable vaccine antigen. 226NKVAS230 and 306TTKKPKD312 are well exposed and possess potential protective properties according to Vaxijen analysis. Our results open the possibility to use these epitopes as suitable anti-*C. difficile* vaccine antigens.

Regulation of *Clostridium difficile* spore germination by the CspA pseudoprotease domain.

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Clostridium difficile is a spore-forming obligate anaerobe that is a leading cause of healthcare-associated infections. *C. difficile* infections begin when its metabolically dormant spores germinate in the gut of susceptible individuals. Binding of bile salt germinants to the Csp family pseudoprotease CspC triggers a proteolytic signaling cascade consisting of the Csp family protease CspB and the cortex hydrolase SleC. Conserved across many of the Clostridia, Csp proteases are subtilisin-like serine proteases that activate pro-SleC by cleaving off its inhibitory pro-peptide. Active SleC degrades the protective cortex layer, allowing spores to resume metabolism and growth. This signaling pathway, however, is differentially regulated in *C. difficile*, since CspC functions both as a germinant receptor and regulator of CspB activity. CspB is also produced as a fusion to a catalytically inactive CspA domain that subsequently undergoes interdomain processing during spore formation. In this study, we investigated the role of the CspA pseudoprotease domain in regulating *C. difficile* spore germination. Mutational analyses revealed that the CspA domain controls CspC germinant receptor levels in mature spores and is required for optimal spore germination, particularly when CspA is fused to the CspB protease. During spore formation, the YabG protease separates these domains, although YabG itself is dispensable for germination. Bioinformatic analyses of Csp family members suggest that the CspC-regulated signaling pathway characterized in *C. difficile* is conserved in related Peptostreptococcaceae family members but not in the Clostridiaceae or Lachnospiraceae. Our results indicate that pseudoproteases play critical roles in regulating *C. difficile* spore germination and highlight that diverse mechanisms control spore germination in the Clostridia.

Role of SpoIVA ATPase Motifs during *Clostridioides difficile* Sporulation.

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The nosocomial pathogen *Clostridioides difficile* is a spore-forming obligate anaerobe that depends on its aerotolerant spore form to transmit infections. Functional spore formation depends on the assembly of a proteinaceous layer known as the coat around the developing spore. In *C. difficile*, coat assembly depends on the conserved spore protein SpoIVA and the clostridial-organism-specific spore protein SipL, which directly interact. Mutations that disrupt their interaction cause the coat to mislocalize and impair spore formation. In *Bacillus subtilis*, SpoIVA is an ATPase that uses ATP hydrolysis to drive its

polymerization around the forespore. Loss of SpoIVA ATPase activity impairs *B. subtilis* SpoIVA encasement of the forespore and activates a quality control mechanism that eliminates these defective cells. Since this mechanism is lacking in *C. difficile*, we tested whether mutations in the *C. difficile* SpoIVA ATPase motifs impact functional spore formation. Disrupting *C. difficile* SpoIVA ATPase motifs resulted in phenotypes that were typically >104-fold less severe than the equivalent mutations in *B. subtilis*. Interestingly, mutation of ATPase motif residues predicted to abrogate SpoIVA binding to ATP decreased the SpoIVA-SipL interaction, whereas mutation of ATPase motif residues predicted to disrupt ATP hydrolysis but maintain ATP binding enhanced the SpoIVA-SipL interaction. When a sipL mutation known to reduce binding to SpoIVA was combined with a spoIVA mutation predicted to prevent SpoIVA binding to ATP, spore formation was severely exacerbated. Since this phenotype is allele specific, our data imply that SipL recognizes the ATP-bound form of SpoIVA and highlight the importance of this interaction for functional *C. difficile* spore formation. **IMPORTANCE** The major pathogen *Clostridioides difficile* depends on its spore form to transmit disease. However, the mechanism by which *C. difficile* assembles spores remains poorly characterized. We previously showed that binding between the spore morphogenetic proteins SpoIVA and SipL regulates assembly of the protective coat layer around the forespore. In this study, we determined that mutations in the *C. difficile* SpoIVA ATPase motifs result in relatively minor defects in spore formation, in contrast with *Bacillus subtilis*. Nevertheless, our data suggest that SipL preferentially recognizes the ATP-bound form of SpoIVA and identify a specific residue in the SipL C-terminal LysM domain that is critical for recognizing the ATP-bound form of SpoIVA. These findings advance our understanding of how SpoIVA-SipL interactions regulate *C. difficile* spore assembly.

Targeting the ATP synthase in bacterial and fungal pathogens: beyond *Mycobacterium tuberculosis*.

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The ATP synthase is a multicomponent enzyme that is largely conserved across the kingdoms of life. In many species the ATP synthase is central in the synthesis of ATP by using the electrochemical proton gradient generated via the electron transport chain. Bacteria inhabit very diverse ecological niches; hence their metabolism to extract nutrients and generation of ATP varies from species to species. Some species are obligate aerobes (e.g., *Mycobacterium tuberculosis*), relying on oxidative phosphorylation for ATP synthesis, whereas others are strict anaerobes (e.g., *Clostridioides difficile*) relying primarily on substrate-level phosphorylation using various fermentative pathways. Yet other species, such as *Staphylococcus aureus* and *Escherichia coli* are facultative anaerobes and can convert energy via both respiratory and fermentative pathways. The metabolic propensity and growth conditions experienced by bacterial species have a great impact on the necessity of a functional ATP synthase for viability. The ATP synthase has been validated as a druggable target with the approval of the ATP synthase inhibitor bedaquiline for treatment of *M. tuberculosis*, an organism in which the ATP synthase is essential for growth. Currently, no ATP synthase inhibitors are in clinical use against non-mycobacterial pathogens. In this review, the physiological functions of the ATP synthase in various bacterial pathogens are discussed in relation to the metabolic pathways utilized for providing energy. The ATP synthase is essential in important pathogenic species that are obligate aerobes, obligate anaerobes and aerotolerant anaerobes, whereas it is dispensable for growth in most facultative anaerobic pathogens. Interference with the ATP synthase in facultative anaerobes has physiological consequences, such as membrane hyperpolarization, which can be exploited for combination therapies. Collectively, the available data indicate that the ATP synthase is an interesting target for development of new antimicrobials beyond *M. tuberculosis*.

Clostridium difficile spore biology: sporulation, germination, and spore structural proteins.

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Clostridium difficile is a Gram-positive, spore-forming obligate anaerobe and a major nosocomial pathogen of worldwide concern. Owing to its strict anaerobic requirements, the infectious and transmissible morphotype is the dormant spore. In susceptible patients, *C. difficile* spores germinate in the colon to form the vegetative cells that initiate *Clostridium difficile* infections (CDI). During CDI, *C. difficile* induces a sporulation pathway that produces more spores; these spores are responsible for the persistence of *C. difficile* in patients and horizontal transmission between hospitalized patients. Although important to the *C. difficile* lifecycle, the *C. difficile* spore proteome is poorly conserved when compared to members of the *Bacillus* genus. Further, recent studies have revealed significant differences between *C. difficile* and *Bacillus subtilis* at the level of sporulation, germination, and spore coat and exosporium morphogenesis. In this review, the regulation of the sporulation and germination pathways and the morphogenesis of the spore coat and exosporium will be discussed.
